

# Structural Reduction of Two Twin-Prime Conjectures and Conditional Resolution under Hypothesis H

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## Abstract

I introduced the Twin-Prime Propagation Conjecture and the Twin-Gap Existence Conjecture in two earlier notes [2, 3]. Here I give a complete structural reduction of both statements. A covering argument shows that the second linear form  $D_n$  is never a lower twin prime after the first pair; consequently the propagation conjecture reduces to a single admissible 6-tuple of linear forms and follows from Schinzel’s Hypothesis H. The revised edition sharpens this in several directions. Under Hypothesis H, *every* gap  $g \equiv 0 \pmod{6}$  occurs infinitely often as the gap of a consecutive propagating pair (Theorem 4.6), by a Chinese-remainder construction that forces consecutiveness. All gap-6 witnesses beyond  $n = 5$  lie in the class  $n \equiv 11 \pmod{30}$ , and gap-6 propagation never iterates: the successor of a gap-6 pair never begins another gap-6 pair (Proposition 4.9). For the gap-existence conjecture the 5-tuple in the gap variable is shown to be admissible for every lower twin  $t > 5$ , with no exceptional set, and its singular series is bounded below by the Hardy–Littlewood quintuplet constant  $10.1318\dots$ ; the conjecture therefore follows from a uniform Bateman–Horn law with  $\gg t^\theta/(\log t)^5$  productive gaps below  $t^\theta$ . The Bateman–Horn constant of the propagating 6-tuple is computed,  $\mathfrak{S} = 51.8959\dots$ , and a heuristic constant  $\kappa = 14.7677\dots$  for the number of propagating consecutive pairs is derived and confirmed against the data to within 3%. The logical relation between the two conjectures is corrected: neither is known to imply the other, both imply the twin-prime conjecture, and both follow from uniform Bateman–Horn. A companion Python certificate (`prove_section6.py`) machine-checks every modular lemma, the admissibility statements, the constructions, the logical reductions, and the computational claims of Section 6.

## Contents

|          |                                                                |           |
|----------|----------------------------------------------------------------|-----------|
| <b>1</b> | <b>The two conjectures</b>                                     | <b>2</b>  |
| <b>2</b> | <b>A covering that kills <math>D_n</math></b>                  | <b>2</b>  |
| <b>3</b> | <b>Both conjectures imply the twin-prime conjecture</b>        | <b>3</b>  |
| <b>4</b> | <b>Resolution of Conjecture 1.1 under Hypothesis H</b>         | <b>4</b>  |
| 4.1      | Every gap occurs . . . . .                                     | 5         |
| 4.2      | Gap-6 propagation never iterates . . . . .                     | 6         |
| 4.3      | The Bateman–Horn asymptotic . . . . .                          | 6         |
| <b>5</b> | <b>Resolution of Conjecture 1.2 under uniform Bateman–Horn</b> | <b>7</b>  |
| <b>6</b> | <b>Status of the claims</b>                                    | <b>9</b>  |
| <b>7</b> | <b>The Python certificate</b>                                  | <b>10</b> |

# 1 The two conjectures

Let  $\mathcal{T}$  denote the strictly increasing sequence of lower twin primes: primes  $p$  for which  $p + 2$  is also prime,

$$3, 5, 11, 17, 29, 41, 59, 71, 101, \dots$$

Write  $p_n$  for the  $n$ th term.

**Conjecture 1.1** (Twin-Prime Propagation). *For consecutive terms  $p_n, p_{n+1}$  set*

$$C_n = p_n + p_{n+1} + 1, \quad D_n = p_n + p_{n+1} + 3. \quad (1)$$

*The pair  $(p_n, p_{n+1})$  propagates if at least one of  $C_n, D_n$  lies in  $\mathcal{T}$ . There are infinitely many such indices  $n$ .*

**Conjecture 1.2** (Twin-Gap Existence). *An even positive integer  $d$  is a productive gap for  $t \in \mathcal{T}$  if*

$$(i) \quad t + d \in \mathcal{T},$$

$$(ii) \quad 2t + d + 1 \in \mathcal{T},$$

$$(iii) \quad \text{at least one of } d - 1, d + 1 \text{ is prime.}$$

*Every sufficiently large  $t \in \mathcal{T}$  admits a productive gap of size  $o(t)$ . If  $d(t)$  is such a gap, the map*

$$G(t) := 2t + d(t) + 1$$

*produces infinitely many twin primes.*

The two constructions coincide on consecutive twins: if  $d = p_{n+1} - p_n$  then  $G(p_n) = C_n$ . Conjecture 1.1 asks that this occur infinitely often along consecutive pairs. Conjecture 1.2 asks for a (possibly non-consecutive) productive  $d = o(t)$  at every large  $t$ .

*Remark 1.3* (Small cases and the threshold). The threshold “sufficiently large” in Conjecture 1.2 cannot be lowered to  $t = 3$ : the conditions  $3 + d \in \mathcal{T}$  and  $7 + d \in \mathcal{T}$  force  $d \equiv 2$  and  $d \equiv 4 \pmod{6}$  respectively (every lower twin  $> 3$  is  $\equiv 5 \pmod{6}$ , Section 2), so  $t = 3$  has no productive gap at all. Every lower twin  $5 \leq t \leq 10^6$  does have one (Section 6), and the data suggest that the threshold is exactly  $t \geq 5$ . Condition (iii) is automatically satisfied by every  $d \equiv 0 \pmod{6}$  with  $6 \leq d \leq 114$ ; the first failure is  $d = 120$ , where  $119 = 7 \cdot 17$  and  $121 = 11^2$ . Consequently (iii) is not a restriction for small gaps, but it is a genuine one in general.

# 2 A covering that kills $D_n$

The disjunction “ $C_n$  or  $D_n$ ” in Conjecture 1.1 is almost redundant.

**Lemma 2.1** (Ternary obstruction). *If  $p, q \in \mathcal{T}$  and  $p, q > 3$ , then  $D := p + q + 3$  is never a lower twin prime.*

*Proof.* Every prime greater than 3 is congruent to  $\pm 1 \pmod{6}$ . A lower twin greater than 3 cannot be  $\equiv 1 \pmod{6}$ , for then the upper member would be  $\equiv 3 \pmod{6}$  and hence divisible by 3. It cannot be  $\equiv 3 \pmod{6}$ , for then it would itself be divisible by 3. Thus every lower twin  $p > 3$  satisfies  $p \equiv 5 \pmod{6}$ , equivalently  $p \equiv 2 \pmod{3}$ . Taking  $p \equiv q \equiv 2 \pmod{3}$  we obtain

$$D + 2 = p + q + 5 \equiv 2 + 2 + 2 \equiv 0 \pmod{3},$$

and  $D + 2 \geq 5 + 5 + 5 = 15 > 3$ . Hence  $D + 2$  is composite, so  $D \notin \mathcal{T}$ . □

The only exception is the first pair  $(p_1, p_2) = (3, 5)$ , for which  $D_1 = 11 \in \mathcal{T}$ . For all  $n \geq 2$ , propagation is exactly the assertion that  $C_n \in \mathcal{T}$ , i.e. that both  $C_n$  and  $C_n + 2 = D_n$  are prime.

**Lemma 2.2** (Residue of  $C_n$ ). *If  $p, q \in \mathcal{T}$  and  $p, q > 3$ , then  $C := p + q + 1 \equiv 5 \pmod{6}$ . In particular there is no obstruction modulo 2 or 3 to  $C \in \mathcal{T}$ .*

*Proof.* From  $p \equiv q \equiv 5 \pmod{6}$  we get  $C \equiv 5 + 5 + 1 \equiv 5 \pmod{6}$ , hence  $C \equiv 2 \pmod{3}$  and  $C + 2 \equiv 1 \pmod{3}$ .  $\square$

**Lemma 2.3** (Gaps between lower twins). *If  $p < q$  are lower twins with  $p > 3$ , then  $q - p \equiv 0 \pmod{6}$ . Moreover, if  $t \in \mathcal{T}$ ,  $t > 3$ , and  $d$  is a productive gap for  $t$ , then  $d \equiv 0 \pmod{6}$ .*

*Proof.* Both  $p$  and  $q$  are  $\equiv 5 \pmod{6}$ . For the second statement apply the first to the pair  $(t, t + d)$ .  $\square$

Thus Conjecture 1.1 is equivalent to the existence of infinitely many consecutive pairs  $p < q$  in  $\mathcal{T}$  for which  $p + q + 1$  and  $p + q + 3$  are both prime, and every gap in sight is a multiple of 6.

### 3 Both conjectures imply the twin-prime conjecture

**Proposition 3.1.** *Conjecture 1.1 implies that there are infinitely many twin primes. Conjecture 1.2 implies the same conclusion.*

*Proof.* A propagating pair produces an element of  $\mathcal{T}$  of size  $\sim 2p_n$ ; infinitely many such pairs therefore yield infinitely many twins. For Conjecture 1.2, take  $t_0 \in \mathcal{T}$  larger than the threshold and set  $t_{k+1} := G(t_k)$ . Then  $t_{k+1} \in \mathcal{T}$  and  $t_{k+1} = 2t_k + d(t_k) + 1 > t_k$ , so  $(t_k)_{k \geq 0}$  is a strictly increasing sequence in  $\mathcal{T}$ .  $\square$

An unconditional proof of either statement would therefore prove the twin-prime conjecture. The Zhang–Maynard–Tao theorem on bounded gaps between primes [9, 6, 7] produces infinitely many prime pairs at distance at most 246, but does not produce a six-dimensional constellation of the shape appearing here, and does not produce gap 2 infinitely often. The parity problem of classical sieves remains. There is no unconditional proof with present technology.

**Proposition 3.2** (Logical relations). *Hypothesis H implies Conjecture 1.1 (Theorem 4.4). The uniform Bateman–Horn hypothesis of Section 5 implies both conjectures. Neither of Conjectures 1.1, 1.2 is known to imply the other.*

*Proof.* The first two assertions are Theorems 4.4 and 5.5 together with the observation that uniform Bateman–Horn implies Bateman–Horn, which implies Hypothesis H for the tuple (2). For the last assertion: Conjecture 1.1 is an infinitely-often statement about *consecutive* pairs, while Conjecture 1.2 is a for-all-large- $t$  statement about gaps that need not be consecutive. A productive gap  $d(t)$  supplied by Conjecture 1.2 is the consecutive gap only if no lower twin lies strictly between  $t$  and  $t + d(t)$ , which Conjecture 1.2 does not assert; in the data the least productive gap is the consecutive gap for fewer than 7% of the lower twins below  $10^6$ . Conversely a consecutive propagating gap need not satisfy (iii) (Remark 1.3), and infinitely many successes say nothing about every large  $t$ .  $\square$

In the first edition of this note Conjecture 1.2 was described as “strictly stronger” than Conjecture 1.1; Proposition 3.2 replaces that claim.

## 4 Resolution of Conjecture 1.1 under Hypothesis H

**Conjecture 4.1** (Schinzel’s Hypothesis H [8]). *Let  $f_1, \dots, f_k \in \mathbb{Z}[x]$  be irreducible polynomials with positive leading coefficients, such that the product  $f_1 \cdots f_k$  has no fixed prime divisor (the system is admissible). Then there are infinitely many integers  $n$  for which every  $f_i(n)$  is prime.*

Consider the six linear forms

$$f_1(n) = n, \quad f_2(n) = n+2, \quad f_3(n) = n+6, \quad f_4(n) = n+8, \quad f_5(n) = 2n+7, \quad f_6(n) = 2n+9. \quad (2)$$

If they are simultaneously prime, then  $n$  and  $n+6$  are lower twins and

$$C = n + (n+6) + 1 = 2n+7$$

is a lower twin, so the pair  $(n, n+6)$  propagates.

**Lemma 4.2** (Admissibility). *The 6-tuple (2) is admissible. Writing  $\nu(q)$  for the number of residues  $r$  modulo  $q$  at which  $\prod f_i(r) \equiv 0$ , one has*

$$\nu(2) = 1, \quad \nu(3) = 2, \quad \nu(5) = 4, \quad \nu(7) = 4, \quad \nu(q) = 6 \text{ for every prime } q \geq 11.$$

*Proof.* It is enough to show that for every prime  $q$  the reductions  $f_i(x) \bmod q$  do not cover  $\mathbb{F}_q$ .

*The prime  $q = 2$ .* Take  $n \equiv 1 \pmod{2}$ . Every  $f_i(n)$  is odd. The only root is  $n \equiv 0$ , so  $\nu(2) = 1$ .

*The prime  $q = 3$ .* Take  $n \equiv 2 \pmod{3}$ . Then  $n \equiv 2, n+2 \equiv 1, n+6 \equiv 2, n+8 \equiv 1, 2n+7 \equiv 2, 2n+9 \equiv 1$ . None vanish; the roots are  $n \equiv 0, 1$ , so  $\nu(3) = 2$ .

*The prime  $q = 5$ .* The roots are  $n \equiv 0, 3, 4, 2, 4, 3$  respectively, i.e. the set  $\{0, 2, 3, 4\}$ , so  $\nu(5) = 4$  and  $n \equiv 1 \pmod{5}$  survives:  $n \equiv 1, n+2 \equiv 3, n+6 \equiv 2, n+8 \equiv 4, 2n+7 \equiv 4, 2n+9 \equiv 1$ .

*The prime  $q = 7$ .* The roots are  $n \equiv 0, 5, 1, 6, 0, 6$ , so  $\nu(7) = 4$  and  $n \equiv 2, 3, 4$  survive.

*Primes  $q \geq 11$ .* Two roots of linear forms  $a_i x + b_i, a_j x + b_j$  coincide modulo  $q$  only if  $q$  divides the minor  $a_i b_j - a_j b_i$ . For the forms (2) the nonzero minors are 2, 3, 4, 5, 6, 7, 8, 9 in absolute value, so for  $q \geq 11$  the six roots are distinct and  $\nu(q) = 6 < q$ . Hence no prime is a fixed divisor of the product  $f_1 \cdots f_6$ .  $\square$

**Lemma 4.3** (Gap 6 forces consecutiveness). *If  $n > 3$  and both  $n, n+6 \in \mathcal{T}$ , then  $n$  and  $n+6$  are consecutive in  $\mathcal{T}$ .*

*Proof.* The only odd integers strictly between  $n$  and  $n+6$  are  $n+2$  and  $n+4$ . Now  $n > 3$  and  $n \in \mathcal{T}$  imply  $n \equiv 2 \pmod{3}$ , so  $n+4 \equiv 0 \pmod{3}$  and  $n+4 \geq 9$ . Thus  $n+4$  is composite, so neither  $n+2$  (whose upper twin would be  $n+4$ ) nor  $n+4$  lies in  $\mathcal{T}$ .  $\square$

**Theorem 4.4** (Conditional resolution of Conjecture 1.1). *Assume Hypothesis H. Then Conjecture 1.1 is true. More precisely, there are infinitely many consecutive pairs in  $\mathcal{T}$  at gap 6 that propagate.*

*Proof.* Lemma 4.2 and Hypothesis H give infinitely many  $n$  for which  $f_1(n), \dots, f_6(n)$  are all prime. For all such  $n > 3$  we have  $n, n+6 \in \mathcal{T}$  and  $2n+7 \in \mathcal{T}$ . Lemma 4.3 says these are consecutive in  $\mathcal{T}$ , and  $C = n + (n+6) + 1 = 2n+7 \in \mathcal{T}$ , so the pair propagates.  $\square$

This is the natural precise form of the “sufficiently strong Hardy–Littlewood” heuristic I used in the earlier notes: it is Hypothesis H for one explicit admissible 6-tuple.

**Lemma 4.5** (The class of gap-6 witnesses). *If  $n > 5$  and  $f_1(n), \dots, f_6(n)$  are all prime, then  $n \equiv 11 \pmod{30}$ .*

*Proof.* By the proof of Lemma 4.2 the only surviving classes are  $n \equiv 1 \pmod{2}$ ,  $n \equiv 2 \pmod{3}$  and  $n \equiv 1 \pmod{5}$ ; for  $n > 5$  none of the six values can equal 2, 3 or 5, so  $n$  must lie in a surviving class. The Chinese remainder theorem gives  $n \equiv 11 \pmod{30}$ .  $\square$

The witnesses below  $10^7$  are  $n = 5, 11, 1481, 116531, 276041, 295871, 388691, \dots$  (62 in all); every one beyond 5 is  $\equiv 11 \pmod{30}$ .

#### 4.1 Every gap occurs

Theorem 4.4 realises only the gap 6. The following strengthening shows that, under Hypothesis H, no gap is missed.

**Theorem 4.6** (Every gap  $g \equiv 0 \pmod{6}$  propagates infinitely often). *Assume Hypothesis H. For every  $g \equiv 0 \pmod{6}$ ,  $g \geq 6$ , there are infinitely many consecutive pairs  $(p, q)$  in  $\mathcal{T}$  with  $q - p = g$  that propagate.*

For a gap  $g$  write

$$\begin{aligned} f_1^{(g)}(n) &= n, & f_2^{(g)}(n) &= n + 2, & f_3^{(g)}(n) &= n + g, \\ f_4^{(g)}(n) &= n + g + 2, & f_5^{(g)}(n) &= 2n + g + 1, & f_6^{(g)}(n) &= 2n + g + 3. \end{aligned} \quad (3)$$

Simultaneous primality gives  $n, n + g \in \mathcal{T}$  and  $C = 2n + g + 1 \in \mathcal{T}$ ; what is missing is consecutiveness.

**Lemma 4.7** (Local admissibility for every gap). *For every  $g \equiv 0 \pmod{6}$  the 6-tuple (3) is admissible. The surviving classes modulo 5 are*

| $g \pmod{30}$          | 0       | 6       | 12         | 18         | 24      |
|------------------------|---------|---------|------------|------------|---------|
| surviving $n \pmod{5}$ | $\{4\}$ | $\{1\}$ | $\{2, 4\}$ | $\{1, 4\}$ | $\{2\}$ |

*Proof.* Modulo 2 take  $n$  odd; since  $g$  is even all six values are odd. Modulo 3 take  $n \equiv 2$ ; since  $g \equiv 0$  the values are 2, 1, 2, 1, 2, 1, none zero. Modulo 5 the roots are 0, 3,  $-g$ ,  $-g - 2$ ,  $-(g + 1)/2$ ,  $-(g + 3)/2$ ; running through the five classes of  $g$  modulo 5 gives the table (for instance  $g \equiv 2$ : roots  $\{0, 3, 3, 1, 1, 0\}$ , survivors  $\{2, 4\}$ ). For  $q \geq 7$  six forms forbid at most six residues and  $q > 6$ .  $\square$

*Proof of Theorem 4.6.* Fix  $g \equiv 0 \pmod{6}$  and put  $J = g/6 - 1$ . By Lemma 2.3 a lower twin  $n + k$  with  $0 < k < g$  and  $n \equiv 5 \pmod{6}$  must have  $k \equiv 0 \pmod{6}$ ; so the only candidates for a lower twin strictly between  $n$  and  $n + g$  are  $n + 6j$ ,  $1 \leq j \leq J$ . For  $g = 6$  there are none and Theorem 4.4 applies, so assume  $g \geq 12$ .

Let  $q_1 < \dots < q_J$  be the  $J$  smallest primes exceeding  $g$ ; then  $q_j \geq 13$ , so  $q_j \nmid 30$ . Choose  $r_{30}$  with  $r_{30} \equiv 1 \pmod{2}$ ,  $r_{30} \equiv 2 \pmod{3}$  and  $r_{30} \pmod{5}$  in the surviving set of Lemma 4.7, and let

$$M = 30 q_1 \cdots q_J, \quad r \equiv r_{30} \pmod{30}, \quad r \equiv -6j \pmod{q_j} \quad (1 \leq j \leq J),$$

by the Chinese remainder theorem. Define the six linear forms in  $m$

$$F_i(m) := f_i^{(g)}(Mm + r) = a_i M m + f_i^{(g)}(r), \quad a_i \in \{1, 2\}.$$

$F_1, \dots, F_6$  is admissible. For  $q \in \{2, 3, 5\}$ ,  $F_i(m) \equiv f_i^{(g)}(r_{30}) \not\equiv 0$  by the choice of  $r_{30}$ . For  $q = q_j$ ,  $F_i(m) \equiv f_i^{(g)}(-6j) \pmod{q_j}$ , and the six values

$$-6j, \quad 2 - 6j, \quad g - 6j, \quad g + 2 - 6j, \quad g + 1 - 12j, \quad g + 3 - 12j$$

are nonzero integers of absolute value less than  $g < q_j$  (the last two are odd, and  $1 \leq j \leq g/6-1$ ), so none vanishes modulo  $q_j$ . For  $q \nmid M$  we have  $q \geq 7$  and  $M$  is invertible modulo  $q$ , so each  $F_i$  has exactly one root and six forms leave a surviving class. Each  $F_i$  is primitive, since  $f_i^{(g)}(r)$  is coprime to  $a_i M$ .

*Consecutiveness.* Hypothesis H gives infinitely many  $m \geq 1$  with all  $F_i(m)$  prime. Put  $n = Mm + r$ . Then  $n, n + g \in \mathcal{T}$ ,  $C = 2n + g + 1 \in \mathcal{T}$ , and for each  $1 \leq j \leq J$  we have  $q_j \mid n + 6j$  with  $n + 6j > M > q_j$ , so  $n + 6j$  is composite. No lower twin lies strictly between  $n$  and  $n + g$ , the pair  $(n, n + g)$  is consecutive in  $\mathcal{T}$ , and it propagates.  $\square$

For  $g = 12$  the construction gives  $M = 390$ ,  $r = 137$ , and the class  $n \equiv 137 \pmod{390}$  contains the witnesses  $n = 29387$  ( $n + 6 = 13 \cdot 2261$ ,  $C = 58787$ ),  $n = 108947$  and  $n = 7403117$  below  $10^7$ . Without the forcing class the data are richer: every gap  $g \equiv 0 \pmod{6}$  with  $g \leq 120$  is realised by a propagating consecutive pair below  $10^7$ , the first examples being

| $g$                    | 6                                | 12                              | 18                           |
|------------------------|----------------------------------|---------------------------------|------------------------------|
| $(p, q) \rightarrow C$ | $(5, 11) \rightarrow 17$         | $(29, 41) \rightarrow 71$       | $(41, 59) \rightarrow 101$   |
| $g$                    | 24                               | 30                              | 36                           |
| $(p, q) \rightarrow C$ | $(5417, 5441) \rightarrow 10859$ | $(2309, 2339) \rightarrow 4649$ | $(311, 347) \rightarrow 659$ |

**Corollary 4.8.** *Under Hypothesis H the set of gaps  $\{p_{n+1} - p_n : n \geq 2, (p_n, p_{n+1}) \text{ propagates}\}$  is exactly  $6\mathbb{Z}_{>0}$ ; unconditionally it is contained in  $6\mathbb{Z}_{>0}$  by Lemma 2.3.*

## 4.2 Gap-6 propagation never iterates

**Proposition 4.9.** *Let  $(n, n + 6)$  be a propagating pair of consecutive lower twins with successor  $C = 2n + 7 \in \mathcal{T}$ . Then  $C + 6 \notin \mathcal{T}$ ; in particular  $(C, C + 6)$  is never a propagating gap-6 pair, and the gap-6 process cannot be iterated even once. Moreover, if  $n > 5$  and  $C'$  is the lower twin following  $C$ , then  $C' - C \equiv 0, 12 \text{ or } 18 \pmod{30}$ .*

*Proof.* For  $n = 5$ :  $C = 17$ , and 23 is prime but  $25 = 5^2$ , so  $23 \notin \mathcal{T}$ ; the next lower twin is 29, at gap 12. For  $n > 5$  Lemma 4.5 gives  $n \equiv 1 \pmod{5}$ , hence  $C = 2n + 7 \equiv 4$  and  $C + 6 \equiv 0 \pmod{5}$  with  $C + 6 > 5$ ; so  $C + 6$  is composite. Equivalently, the 10-tuple obtained by adjoining  $2n + 13, 2n + 15, 4n + 21, 4n + 23$  to (2) has the fixed divisor 5: its roots modulo 5 cover  $\mathbb{F}_5$ . For the last claim,  $C' \in \mathcal{T}$  with  $C' > 5$  requires  $C' \not\equiv 0, 3 \pmod{5}$ , so  $C' - C \not\equiv 1, 4 \pmod{5}$ ; combined with  $C' - C \equiv 0 \pmod{6}$  this leaves the classes 0, 12, 18 modulo 30.  $\square$

Among the gap-6 witnesses  $5 < n \leq 10^7$  whose successor  $C$  also lies below  $10^7$  (37 of them) the following gap is  $\equiv 0, 12, 18 \pmod{30}$  in 13, 11, 13 cases respectively, never 6 or 24.

## 4.3 The Bateman–Horn asymptotic

The same argument with the Bateman–Horn conjecture [1] supplies the asymptotic. The singular series

$$\mathfrak{S} = \prod_q \left(1 - \frac{1}{q}\right)^{-6} \left(1 - \frac{\nu(q)}{q}\right) \quad (4)$$

is positive by Lemma 4.2, and Bateman–Horn predicts

$$\#\{n \leq X : f_1(n), \dots, f_6(n) \text{ all prime}\} \sim \mathfrak{S} \int_2^X \frac{dt}{\prod_{i=1}^6 \log f_i(t)} \asymp \frac{X}{(\log X)^6}. \quad (5)$$

(The leading coefficients of  $f_5, f_6$  enter only through  $\nu(2)$ ; the first edition carried a spurious factor  $\prod(\text{lead } f_i)^{-1}$ .) With the values of Lemma 4.2,

$$\mathfrak{S} = 32 \cdot \frac{243}{64} \cdot \frac{3125}{4096} \cdot \frac{50421}{46656} \cdot \prod_{q \geq 11} \left(1 - \frac{1}{q}\right)^{-6} \left(1 - \frac{6}{q}\right) = 51.8959 \dots$$

The product over  $q \geq 11$  is the tail of the Hardy–Littlewood sextuplet constant 17.2986... (the tuple  $(0, 4, 6, 10, 12, 16)$  has  $\nu(7) = 6$ ), so  $\mathfrak{S} = 3 \times 17.2986 \dots$  exactly. The prediction (5) gives 51.7 witnesses below  $10^7$  and 112.6 below  $4 \cdot 10^7$ ; the actual counts are 62 and 126, consistent with Poisson fluctuation. In particular the count of propagating gap-6 pairs up to  $X$  is  $\asymp X/(\log X)^6 \rightarrow \infty$ , and by the proof of Theorem 4.6 the same holds, with its own positive constant, for every gap  $g \equiv 0 \pmod{6}$ .

*Remark 4.10* (Corrected local heuristic). In the original notes I treated  $C_n$  and  $D_n$  as exchangeable and summed  $\int dt/(\log t)^3$ . By Lemma 2.1 the  $D_n$ -branch contributes only the single pair  $(3, 5)$ . Conditioning on a consecutive pair at height  $x$ , one asks that a 2-tuple of size  $\sim 2x$  be twin, which has probability  $\asymp 1/(\log x)^2$ . The twin-counting measure is  $\asymp dt/(\log t)^2$ , so the expected number of successes up to  $X$  is  $\asymp \int_3^X dt/(\log t)^4$ , which still diverges. The empirical fit  $6.175(\ln p)^{-2.657}(\ln \ln p)^{2.258}$  of the earlier notes is a regression on a finite range; the Hardy–Littlewood prediction for the success rate is  $\sim \kappa_{\text{loc}}/(\log 2x)^2$  with the explicit constant below.

**Heuristic 4.11** (The constant  $\kappa$ ). Let  $(p, p')$  be a consecutive pair in  $\mathcal{T}$  with  $p > 5$ , at height  $x$ , and  $C = p + p' + 1$ . Then

$$\Pr(C \in \mathcal{T}) \approx \frac{\kappa_{\text{loc}}}{(\log 2x)^2}, \quad \kappa_{\text{loc}} = 6(2C_2) \prod_{\ell \geq 5} \left(1 + \frac{8}{(\ell - 2)^3}\right) = 11.1849 \dots,$$

where  $2C_2 = 2 \prod_{\ell \geq 3} (1 - (\ell - 1)^{-2}) = 1.32032 \dots$  is the twin-prime constant, and the number of propagating consecutive pairs with  $p_n \leq X$  is

$$\approx \kappa \int_5^X \frac{dt}{(\log t)^2 \log(2t + 1) \log(2t + 3)}, \quad \kappa = \kappa_{\text{loc}} \cdot 2C_2 = 14.7677 \dots$$

*Derivation.* For a random integer  $m \approx y$  the Hardy–Littlewood heuristic [4] gives  $\Pr(m, m + 2 \text{ prime}) \approx (\log y)^{-2} \prod_{\ell} \Pr(m \not\equiv 0, -2 \pmod{\ell}) (1 - 1/\ell)^{-2}$ , and for uniformly distributed  $m$  the product equals  $2C_2$ . Here  $C$  is not uniform. Modulo 2 and 3 it is fixed,  $C \equiv 5 \pmod{6}$  (Lemma 2.2), which contributes the factor  $(1/2)^{-2}(2/3)^{-2} = 9$  in place of  $2 \cdot \frac{3}{4} = \frac{3}{2}$ ; the ratio is 6. For a prime  $\ell \geq 5$  both  $p$  and  $p'$  lie in  $A_{\ell} = \mathbb{F}_{\ell} \setminus \{0, -2\}$ , and to leading order the residues of consecutive lower twins are independent and uniform on  $A_{\ell}$  (the biases of Lemke Oliver–Soundararajan [5] are of relative size  $O(\log \log x / \log x)$ ). The number of pairs  $(a, b) \in A_{\ell}^2$  with  $a + b + 1 \equiv c$  is  $\ell - 4$  for each of the two bad values  $c \in \{0, -2\}$ , because  $\{0, -2\} \cap \{c, c + 2\} = \emptyset$  when  $\ell \geq 5$ . Hence  $\Pr(C \in A_{\ell}) = 1 - 2(\ell - 4)/(\ell - 2)^2$ , and the ratio to the uniform value  $1 - 2/\ell$  is

$$\frac{((\ell - 2)^2 - 2(\ell - 4)) \ell}{(\ell - 2)^3} = \frac{(\ell^2 - 6\ell + 12) \ell}{(\ell - 2)^3} = 1 + \frac{8}{(\ell - 2)^3}.$$

Multiplying the corrections into  $2C_2$  gives  $\kappa_{\text{loc}}$ ; integrating against the density  $2C_2 dt/(\log t)^2$  of  $\mathcal{T}$  gives  $\kappa$ .  $\square$

The data agree with Heuristic 4.11 to within a few percent: the number of propagating consecutive pairs with  $p_n \leq 10^7$  is 2827 against a prediction of 2759.5, and with  $p_n \leq 4 \cdot 10^7$  it is 7937 against 7719.1. In dyadic blocks  $[x, 2x)$  with  $x \geq 6 \cdot 10^5$  the empirical value of (success rate)  $\times (\log 2x)^2$  is 11.6, 10.8, 11.2, 11.1, against  $\kappa_{\text{loc}} = 11.18$ .

## 5 Resolution of Conjecture 1.2 under uniform Bateman–Horn

Fix  $t \in \mathcal{T}$  with  $t > 3$ , so  $t \equiv 5 \pmod{6}$ . Productive even  $d$  with the side-condition that  $d - 1$  is prime correspond to simultaneous primality of the five linear forms in  $d$

$$g_1(d) = d + t, \quad g_2(d) = d + t + 2, \quad g_3(d) = d + 2t + 1, \quad g_4(d) = d + 2t + 3, \quad g_5(d) = d - 1. \quad (6)$$

The alternative  $g_5^*(d) = d + 1$  covers the other clause of condition (iii). If  $g_1(d), g_2(d)$  are both prime and exceed 3 then  $d \equiv 0 \pmod{6}$  automatically (Lemma 2.3).

**Lemma 5.1** (No small-prime obstruction, uniformly in  $t$ ). *Let  $t > 3$ ,  $t \equiv 5 \pmod{6}$ , and  $d \equiv 0 \pmod{6}$ . Then none of  $g_1(d), \dots, g_5(d)$ ,  $g_5^*(d)$  is divisible by 2 or 3.*

*Proof.* The integer  $d$  is even and  $t$  is odd, so all forms are odd. Modulo 3:  $d \equiv 0$  and  $t \equiv 2$ , so  $d + t \equiv 2$ ,  $d + t + 2 \equiv 1$ ,  $d + 2t + 1 \equiv 2$ ,  $d + 2t + 3 \equiv 1$ ,  $d - 1 \equiv 2$ ,  $d + 1 \equiv 1$ . None vanish.  $\square$

**Lemma 5.2** (Admissibility for every  $t > 5$ ). *For every  $t \in \mathcal{T}$  with  $t > 5$  the 5-tuple (6) is admissible. For  $t = 5$  it is not (5 is a fixed divisor), but the starred tuple  $(g_1, \dots, g_4, g_5^*)$  is. The starred tuple is inadmissible for every  $t \equiv 3 \pmod{5}$ .*

*Proof.* Primes 2 and 3 are handled by Lemma 5.1; for  $q \geq 7$  five forms forbid at most five residues and  $q > 5$ . It remains to examine  $q = 5$ , where the roots are  $-t, -t - 2, -2t - 1, -2t - 3$  and 1 (resp.  $-1$  for the starred tuple). Since  $t > 5$  is prime,  $t \not\equiv 0 \pmod{5}$ , and the four cases are

| $t \pmod{5}$ | roots of $g_1..g_4$ | roots with $g_5$ | survivors, $g_5$ | survivors, $g_5^*$ |
|--------------|---------------------|------------------|------------------|--------------------|
| 1            | $\{4, 2, 2, 0\}$    | $\{0, 1, 2, 4\}$ | $\{3\}$          | $\{1, 3\}$         |
| 2            | $\{3, 1, 0, 3\}$    | $\{0, 1, 3\}$    | $\{2, 4\}$       | $\{2\}$            |
| 3            | $\{2, 0, 3, 1\}$    | $\{0, 1, 2, 3\}$ | $\{4\}$          | $\emptyset$        |
| 4            | $\{1, 4, 1, 4\}$    | $\{1, 4\}$       | $\{0, 2, 3\}$    | $\{0, 2, 3\}$      |

In every row the tuple with  $g_5$  has a survivor, and for  $t \equiv 3$  the starred tuple has none. For  $t \equiv 0 \pmod{5}$  the roots of  $g_1, \dots, g_4$  are  $\{0, 3, 4, 2\}$ ; adjoining 1 covers  $\mathbb{F}_5$ , adjoining  $-1 \equiv 4$  leaves the survivor 1.  $\square$

The first edition stated admissibility only “outside a thin set” of  $t$ ; Lemma 5.2 shows that no exceptional set exists.

**Lemma 5.3** (Uniform lower bound for the singular series). *For  $t \in \mathcal{T}$ ,  $t > 5$ , let  $\nu_t(q)$  be the number of roots of  $\prod_{i \leq 5} g_i$  modulo  $q$  and*

$$\mathfrak{S}(t) = \prod_q \left(1 - \frac{1}{q}\right)^{-5} \left(1 - \frac{\nu_t(q)}{q}\right).$$

*Then  $\mathfrak{S}(t) \geq \mathfrak{S}_{\min}$  for every such  $t$ , where*

$$\mathfrak{S}_{\min} = \prod_q \left(1 - \frac{1}{q}\right)^{-5} \left(1 - \frac{\nu^*(q)}{q}\right), \quad \nu^*(2) = 1, \nu^*(3) = 2, \nu^*(5) = 4, \nu^*(q) = 5 \ (q \geq 7),$$

*is the Hardy–Littlewood quintuplet constant,  $\mathfrak{S}_{\min} = 10.1318\dots$*

*Proof.* Each local factor  $(1 - 1/q)^{-5}(1 - \nu/q)$  is strictly decreasing in  $\nu$ . Modulo 2 all five forms reduce to  $d + 1$ , so  $\nu_t(2) = 1$ ; modulo 3 the roots are  $\{1, 2\}$  by Lemma 5.1, so  $\nu_t(3) = 2$ ; modulo 5 Lemma 5.2 gives  $\nu_t(5) \leq 4$ ; and trivially  $\nu_t(q) \leq 5$ . Hence every local factor of  $\mathfrak{S}(t)$  is at least the corresponding factor of  $\mathfrak{S}_{\min}$ . The tuple  $(0, 2, 6, 8, 12)$  has exactly the root counts  $\nu^*$ , so  $\mathfrak{S}_{\min}$  is its Hardy–Littlewood constant.  $\square$

Concretely  $\mathfrak{S}(11) = 22.80$ ,  $\mathfrak{S}(17) = 21.71$ ,  $\mathfrak{S}(29) = 47.35$ ,  $\mathfrak{S}(101) = 15.99$ ,  $\mathfrak{S}(1019) = 78.46$ : coincidences among the roots (which occur when  $q \mid (t - 1)(t + 1)(t + 2)(t + 3)$ ) only increase the series.

**Conjecture 5.4** (Uniform Bateman–Horn hypothesis). *For an admissible family of  $k$  linear forms  $h_1, \dots, h_k$  in one variable, with coefficients of absolute value at most  $X$  and singular series  $\mathfrak{S}$ ,*

$$\#\{n \leq Y : h_1(n), \dots, h_k(n) \text{ all prime}\} = (\mathfrak{S} + o(1)) \int_2^Y \frac{du}{\prod_i \log |h_i(u)|},$$

*uniformly for  $Y \geq X^\theta$  with any fixed  $\theta > 0$ , as  $X \rightarrow \infty$ .*

**Theorem 5.5** (Conditional resolution of Conjecture 1.2). *Assume Conjecture 5.4 for 5-tuples of linear forms. Then for every fixed  $0 < \theta < 1$ , every sufficiently large  $t \in \mathcal{T}$  has at least*

$$(\mathfrak{S}_{\min} + o(1)) \frac{t^\theta}{\theta (\log t)^5}$$

*productive gaps of size at most  $t^\theta$ . In particular Conjecture 1.2 holds, even with the stronger bound  $d(t) = O_\theta(t^\theta)$ .*

*Proof.* Apply Conjecture 5.4 to  $g_1, \dots, g_5$  in the variable  $d \leq Y := t^\theta$ ; the coefficients are  $O(t)$ . Lemma 5.2 gives admissibility for  $t > 5$  and Lemma 5.3 gives  $\mathfrak{S}(t) \geq \mathfrak{S}_{\min}$ . For  $2 \leq u \leq Y$  one has  $\log g_i(u) \leq \log(3t)$  for  $i \leq 4$  and  $\log g_5(u) \leq \log Y = \theta \log t$ , so the integral is at least  $(1 + o(1)) Y / (\theta (\log t)^5)$ . Every  $d$  counted has  $g_1(d), g_2(d)$  prime and  $> 3$ , hence  $d \equiv 0 \pmod{6}$ ,  $d + t \in \mathcal{T}$ ,  $2t + d + 1 \in \mathcal{T}$  and  $d - 1$  prime:  $d$  is productive, of size  $\leq t^\theta = o(t)$ .  $\square$

**Conjecture 5.6** (Least productive gap). *For  $t \in \mathcal{T}$ ,  $t \geq 5$ , let  $d_{\min}(t)$  be the least productive gap. Then  $d_{\min}(t)$  exists for every  $t \geq 5$ , and for every  $\varepsilon > 0$ ,*

$$d_{\min}(t) \ll_\varepsilon (\log t)^{5+\varepsilon}.$$

The exponent is what the Bateman–Horn density suggests: by Lemma 5.3 the productive  $d \leq Y$  have density  $\asymp 1/((\log t)^4 \log Y)$ , so the first hit is typically of size  $(\log t)^4 \log \log t$ , and the maximal first hit over the  $\asymp X/(\log X)^2$  lower twins  $t \leq X$  is, on a Cramér-type model, of size  $(\log X)^5 \log \log X$ . In the data every lower twin  $5 \leq t \leq 10^6$  has a productive gap; the largest least gap is  $d_{\min}(719351) = 63318$ , and  $\max_{11 \leq t \leq 10^6} d_{\min}(t)/(\log t)^5 = 0.2075$ , attained at  $t = 7331$ .

Along the orbit  $t_{k+1} = G(t_k)$  one has  $t_{k+1} \sim 2t_k$ , so  $t_k \asymp 2^k t_0$ . This is the generative map of the earlier notes; under Theorem 5.5 it is defined at every step, and with least productive gaps it runs from 5 as

$$5 \rightarrow 17 \rightarrow 59 \rightarrow 197 \rightarrow 617 \rightarrow 1277 \rightarrow 2729 \rightarrow 5519 \rightarrow 11171 \rightarrow 25799 \rightarrow 52859 \rightarrow \dots$$

## 6 Status of the claims

The following table records the status of every statement used to resolve the two conjectures. The numbering of the rows matches the eight certificates of the companion script.

| Claim                                                                                           | Status                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $D_n \in \mathcal{T}$ for $n \geq 2$                                                          | False (Lemma 2.1). Conjecture 1.1 reduces to $C_n \in \mathcal{T}$ .                                                                                                                                     |
| 2 Conjecture 1.1, unconditionally                                                               | Open. Implies the twin-prime conjecture (Proposition 3.1).                                                                                                                                               |
| 3 Conjecture 1.1 under Hypothesis H                                                             | True (Theorem 4.4), via the admissible 6-tuple (2); every gap $g \equiv 0 \pmod{6}$ occurs infinitely often (Theorem 4.6); witnesses $n > 5$ lie in $11 \pmod{30}$ (Lemma 4.5).                          |
| 4 Gap-6 propagation iterates                                                                    | False, unconditionally (Proposition 4.9); the following gap is $\equiv 0, 12, 18 \pmod{30}$ .                                                                                                            |
| 5 Conjecture 1.2, unconditionally                                                               | Open. Not comparable with Conjecture 1.1 by any known implication (Proposition 3.2); $t = 3$ has no productive gap; every $5 \leq t \leq 10^6$ has one.                                                  |
| 6 Conjecture 1.2 under uniform Bateman–Horn                                                     | True (Theorem 5.5), with $\gg t^\theta/(\log t)^5$ gaps below $t^\theta$ ; the 5-tuple is admissible for every $t > 5$ and $\mathfrak{S}(t) \geq 10.1318$ .                                              |
| 7 Conjecture 1.1 or 1.2 $\Rightarrow$ infinitely many twins                                     | True (Proposition 3.1).                                                                                                                                                                                  |
| 8 Computational support ( $p_n \leq 10^7$ by default; $p_n \leq 10^{13}$ in the original notes) | Consistent with both conjectures, with the Bateman–Horn constant $\mathfrak{S} = 51.8959$ for the gap-6 tuple, and with $\kappa = 14.7677$ for the number of propagating pairs (ratio 1.02 to the data). |

An unconditional proof of either conjecture would be a proof of the twin-prime conjecture by a new route, and would require an idea that bypasses the parity barrier for a six-dimensional prime constellation. That idea is not supplied here. The theorems above are the resolution that the statements actually admit: Conjecture 1.1 is Hypothesis H for one explicit admissible tuple (indeed for a family of tuples, one for each gap); Conjecture 1.2 is a uniform short-interval form of the same law with a singular series bounded below by an absolute constant.

## 7 The Python certificate

The companion script `prove_section6.py` machine-checks every row of the table in Section 6. The modular lemmas are finite residue-class enumerations and are therefore complete proofs. Admissibility of a  $k$ -tuple of primitive linear forms is decided by inspecting the primes  $q \leq k$  exhaustively and invoking the pigeonhole principle for  $q > k$ ; the root counts  $\nu(q)$  are computed exactly below the largest  $2 \times 2$  minor of the coefficient matrix and are equal to  $k$  above it. The construction of Theorem 4.6 is executed for  $12 \leq g \leq 60$ : the script builds  $M$ ,  $r$  and the forms  $F_i$ , verifies admissibility, verifies that  $q_j \mid r + 6j$ , and exhibits witnesses in the class for  $g = 12$ . Proposition 4.9 is checked both by the covering of  $\mathbb{F}_5$  by the 10-tuple and against the data. The singular series  $\mathfrak{S}$ ,  $\mathfrak{S}_{\min}$  and the constants  $2C_2$ ,  $\kappa_{\text{loc}}$ ,  $\kappa$  are evaluated from their Euler products over the primes up to  $10^6$  (relative tail below  $2 \cdot 10^{-6}$ ), and the Bateman–Horn integrals by Simpson’s rule in the variable  $\log t$ ; nothing is fitted. The “open” claims are certified by exhibiting the reduction to the twin-prime conjecture together with the observation that no finite computation decides an infinitary statement of this type. The default run uses  $p_n \leq 10^7$  (sieve to  $2 \cdot 10^7 + 4 \cdot 10^5$ ) and finishes in a few seconds; `-limit` raises the range.

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